Comparative Co-Presence Network Analysis of the New Testament Gospels and the Qur'an

Erin Brzusek

Abstract

This study seeks to analyze textual data from the four New Testament Gospels (Matthew, Mark, Luke, and John) and the Surahs of the Quran to construct and compare co-presence networks. The goal of this project is to evaluate how the distinct literary styles of chronological narratives and thematic discourses alter the structure of the resulting networks. Because the texts differ significantly in length and structure, the methodology requires data normalization, clustering coefficient comparisons, and degree distribution analysis to quantify structural similarities and differences.

1 Introduction

Practitioners of Abrahamic religions make up over half the world's population [1], and the holy texts of these faiths serve as the cornerstone for their respective beliefs and cultural traditions. Despite historical and contemporary contention between Christianity and Islam, the New Testament Gospels and the Qur'an share a deeply interconnected lineage stemming from Abraham. Shared figures bridge them conceptually, but the texts themselves are structured in fundamentally different ways. The four New Testament Gospels (Matthew, Mark, Luke, and John) are framed primarily as chronological narratives, whereas the Surahs of the Qur'an are formatted as a largely thematic discourse.

This study analyzes textual data from both the Bible and Qur'an to construct, visualize, and compare their underlying character networks. Specifically, this study evaluates how these distinct literary styles alter the resulting graph topology among shared entities. Because the texts differ significantly in length, scope, and structural organization, the methodology incorporates data normalization, clustering coefficient comparisons, and degree distribution analysis to quantify structural similarities and differences. Data for this study is drawn from the BibleData repository for the Gospels [1] and the Sahih International English translation via the Qur'an API [2].

This paper seeks to address three central research questions:

- Does a thematic text structure exhibit different network characteristics compared to a chronological narrative?
- How do the centrality metrics of shared characters shift between the two texts?
- Do both networks exhibit power-law degree distributions despite their fundamental structural differences?

Ultimately, this comparative network analysis serves to show that the shared principal figures rise above the structural settings of their respective texts. Beyond religious and literary divides, they represent a shared foundation of human hope and eternal faith.

2 Literature Review

In attempting to analyze textual data, network analytics provides an empirical perspective to the task. Rather than characters being classified as "protagonists", their betweeness centrality can be calculated and visualized.

Social networks imply an interaction between two characters. A co-presence network emphasizes the relationship between characters rather than requiring explicit conversations. It leverages the literary style of a chronological

[1] According to World Population Review, 32% of the global population are Christians, 25% Muslims and 0.2% Jewish

narrative. Implicit relationships are also important.

Religious texts written in ancient languages present a problem for modern natural language processing tools. It's hard to identify who a `"PERSON"` might be when they're given multiple names in the span of the same text. Additionally, determining a threshold for what constitutes an implicit relationship between characters is challenging [3]. This also introduces the deliniation between social, co-presence, and co-occurence networks.

Topics in social network analytics: degree or valency, centrality metrics, clustering coefficients, diameter, degree, density, and ego networks [4]. These metrics transform qualitative relationships between characters into mathematical terms ripe for comparison.

SNA done in the context of Luke's gospel [5].

Analyzed all four gospels, unsurprisingly Jesus is the hub for all four [6]

3 Methodology

Utilizing data from [1] and [2]

Explain translation vs. transliteration: used english Sahih International translation

How to define textual proximity for edges? - the bibledata already defines relationships within verses - Qur'an data preprocessing was set up to mirror that, co-presence established within each verse

Compute network-level metrics

Measure clustering coefficients

Test degree distributions (power-law fitting)

Centralities

Spearman Rank Correlations

Top-K centrality shifts

Community detection? node embeddings?

Visualizations

3.1 Network Metrics

To evaluate structural importance, we measure **degree centrality** and **betweenness centrality**:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

4 Results

5 Discussion

6 Conclusion

7 References

[1] "BibleData." Accessed: Jul. 23, 2026. [Online]. Available: https://www.kaggle.com/datasets/bradystephenson/bibledata

[2] I. Network, “API Documentation · Al Quran Cloud.” Accessed: Jul. 23, 2026. [Online]. Available: https://alquran.cloud

[3] J. Drpinghaus, “Social Network Analysis and Co-Occurrence: Identifying the Gaps,” 2022.

[4] S. E. Massey, “Social network analysis of the biblical Moses,” *Applied Network Science*, vol. 1, no. 1, p. 13, Nov. 2016, doi: 10.1007/s41109-016-0012-1.

[5] J. Dörpinghaus, “A Structural Analysis of ‘In’ and ‘Out’ in Luke’s,” 2026.

[6] M. Grandjean, “Data Visualization: Mapping the Character Network of the Four Gospels | Martin Grandjean.” Jul. 2013. Accessed: Jul. 23, 2026. [Online]. Available: https://www.martingrandjean.ch/data-visualization-network-gospels/